PHYS 400 · SENIOR RESEARCH PROJECT

Quantifying Information in Gravitational-Wave Observations

How many bits does a black-hole merger reveal?

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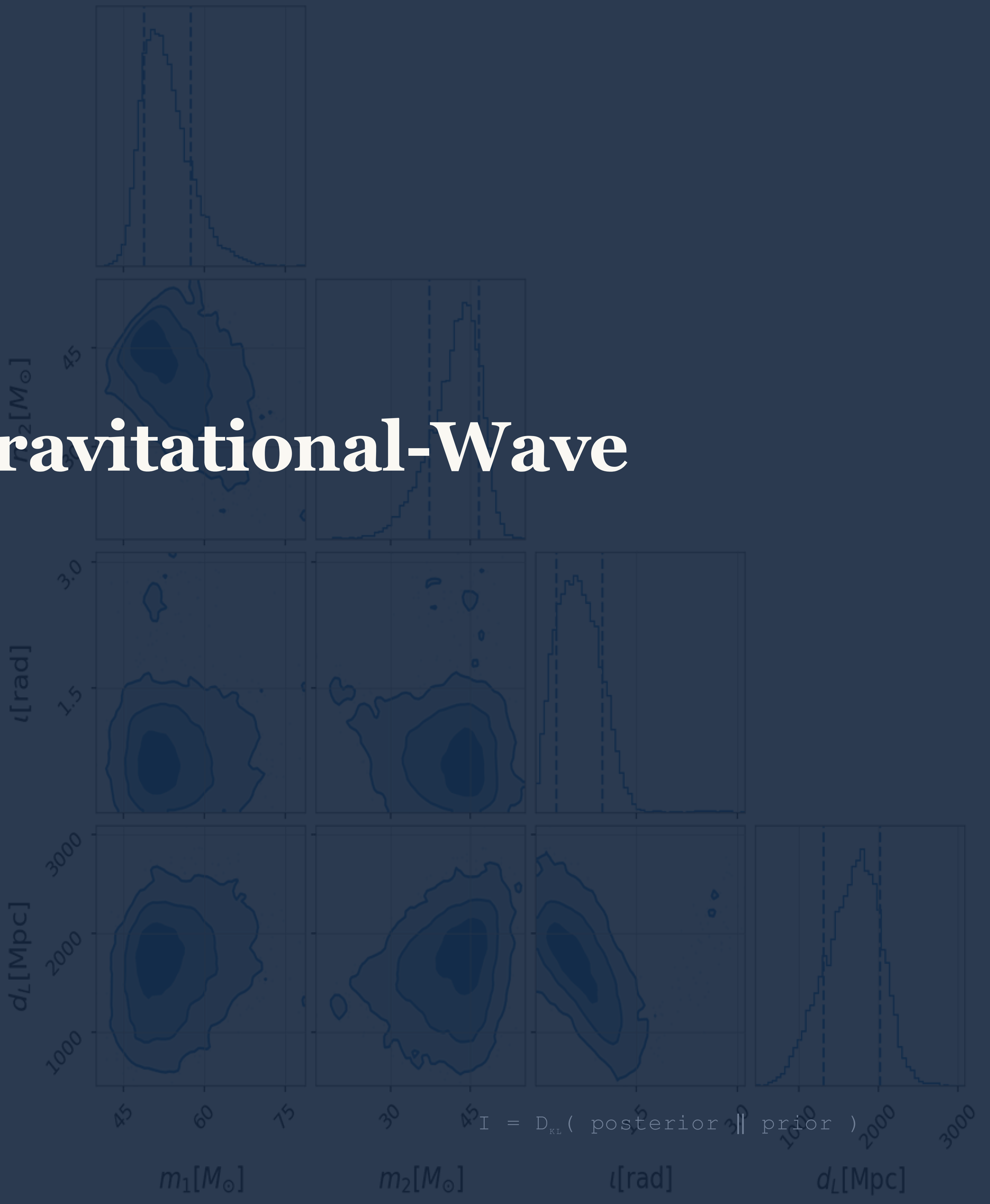


$I = D_{KL}(\text{posterior} \parallel \text{prior})$

Outline

Motivation

Why measure information in GW?

Introduction

Bayesian framework & KL divergence

Method

Group-KLD hybrid prior, estimators

Results

35 events scaling laws

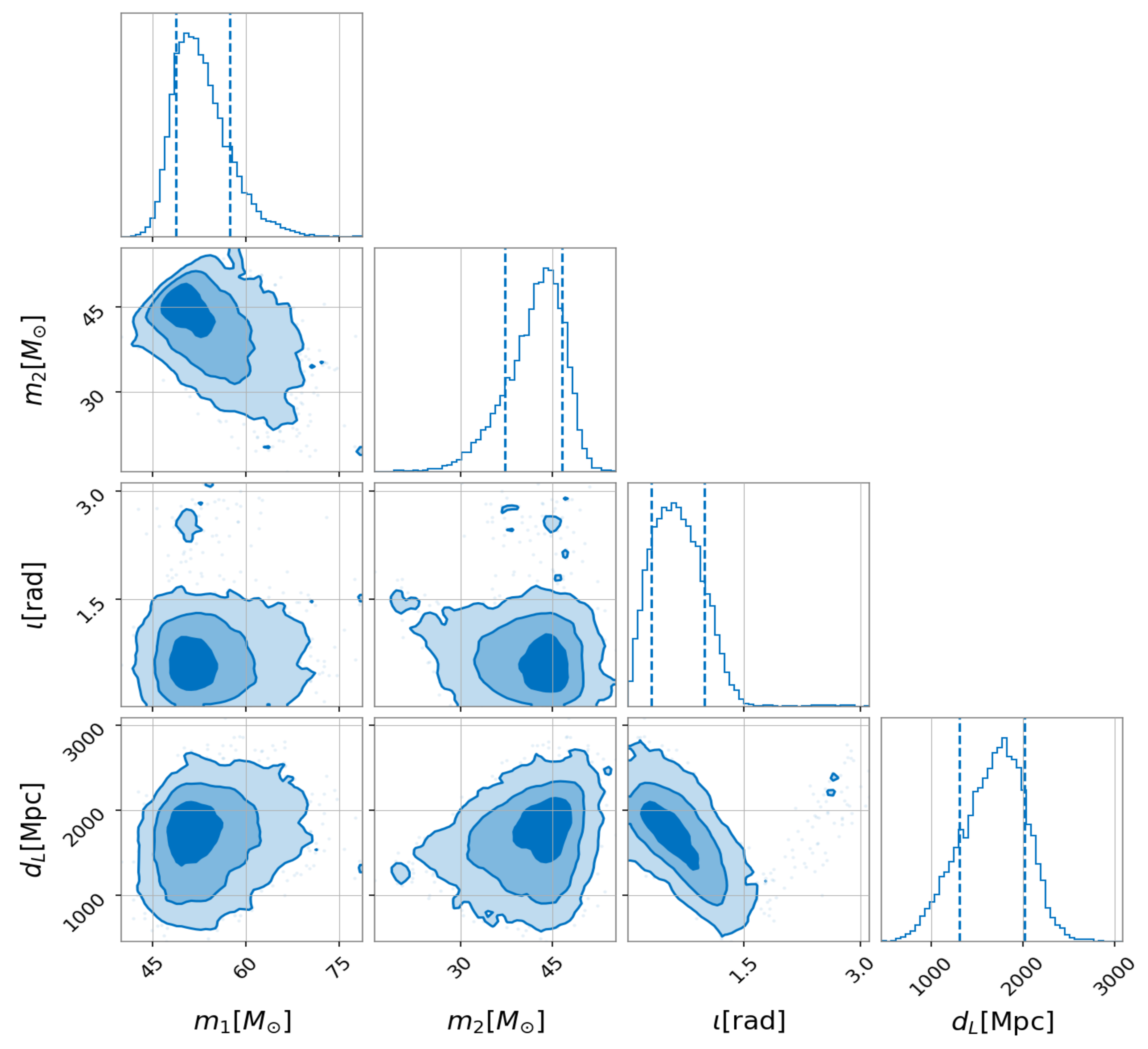
Summary

How much information we obtain from observations ?

Each event include physical information about source.

Which parameters (like mass or spin) give us the most information?

How good are our current tools? (LIGO-Virgo-KAGRA network)
This shows the true efficiency of present observatories.



Posterior for a real BBH event — masses, inclination, distance and their correlations.

Information is the divergence from prior to posterior

$$I = D_{KL}(p(\theta|s) \parallel p^0(\theta)) = \int p(\theta|s) \log_2 [p(\theta|s) / p^0(\theta)]\, d\theta$$

POSTERIOR

$p(\theta|s)$ — what we find about the source after seeing the signal *s*.

PRIOR

$p^0(\theta)$ — what we assumed *before* about the parameters.

UNITS

Log base 2 ⇒ the answer is in **bits** of gained knowledge.

Θ

Source parameters — the 15 physical parameter of BBH.

S

Observed gravitational-wave signal.

Fifteen physical parameters describe a binary black hole

MASSES · 2

m_1, m_2

source-frame component masses

SPINS · 4

$a_1, a_2, \mathrm{tilt}_1, \mathrm{tilt}_2$

magnitudes & tilt angles

SPIN GEOMETRY · 2

$\varphi_{12}, \varphi JL$

azimuthal spin angles

DISTANCE · 1

d_L

luminosity distance

ORIENTATION · 2

θ_{JN}, ψ

inclination & polarization

SKY POSITION · 2

azimuth, zenith

isometric to RA, Dec

TIME · 1

t_c

coalescence time

PHASE · 1

φ_{c1}

reference phase

Same 15-parameter convention as Flanagan & Hughes (1998); posteriors from the `C01:IMRPhenomXPHM` waveform model.

Flanagan & Hughes, 1998

The classic analytic estimate for GW150914-like systems:

≈41.5 bit

$$I = \tfrac{1}{2} N_{\text{param}} \log_2\left(1 + \mathrm{SNR}^2/N_{\text{param}}\right)$$

SNR = signal-to-noise ratio ·

N_{param} = effective number of parameters

RESTS ON TWO ASSUMPTIONS

1. **Gaussian posterior**
   Exact only in the high signal-to-noise limit.
2. **Linear signal manifold**
   The waveform depends linearly to the parameters.

The authors themselves call it "a rather crude approximation." Real posteriors are curved and non-Gaussian — so we compute the KL divergence directly.

35 binary black holes

- Parameter-estimation samples from the **IGWN / GWOSC GWTC data release.**
- GW150914 (GWTC-2.1) plus 34 **GWTC-3** events, read straight from the public `HDF5` products.

35
BBH EVENTS

15
PARAMETERS

10^5
SAMPLES / EVENT

PART I

Estimating KL Divergence in 15 Dimensions

Five estimators, two data traps, and the method that finally scaled to every event.

The estimators (1 Dimensional)

ESTIMATOR	IDEA	GW150914	VERDICT
Kernel Density Estimation (KDE)	Smooth at each data point and sums them to estimate a continuous probability density.	≈ 37.4 bit	OK
Multivariate Gaussian	Assume posterior gaussian	≈ 41.3 bit	high-SNR
k-NN (Wang / Pérez-Cruz)	nearest-neighbour ratios	≈ 39 bit	OK
Histogram (n-D bins)	Direct density	≈ 41 bit	OK

Why don't we compute the one-dimensional KL divergence and sum for all parameters?

A lower bound, and a capacity ceiling

!!! Sum of 1D Marginal Kullback–Leibler divergence ≤ 15D Joint Kullback–Leibler divergence !!!

DECOMPOSITION · LOWER BOUND

$$I_{\text{joint}} = \Sigma_i I_{\text{marg}}(i) + \text{TC}$$

- Total correlation **TC ≥ 0** always.
- The sum of **1-D marginal** KL divergences is a guaranteed lower bound on the true joint information .

Four ways to estimate a 15-D density

NEAREST-NEIGHBOUR

k-NN

Wang / Pérez-Cruz

- Density from neighbour distances.
- **Fully non-parametric.**
- No model assumed.

25.9 → 104.9 bit

Never converges.

PARAMETRIC

Multivariate Gaussian

Covariance log-det

- Simple to calculate.
- **Assumes posteriors are Gaussian Distributions.**
- A result close to the theoretical prediction.

≈ 41.2 bit

Condition number $\sim 10^9$

SMOOTH

KDE

Scott / Silverman

- Smooth kernels over each low-D block.
- **Stable** where samples are dense.
- Takes week to compute.

Impossible to allocate

BINNING

np.histogramdd

direct n-D grid

- Count samples in a full grid.
- Just **$5^{15} \approx 3\times10^{10}$ cells** at 5 bins.

240 GB RAM

Impossible to allocate

Their shared problem is **dimensionality.**

Our estimators stand on a published benchmark

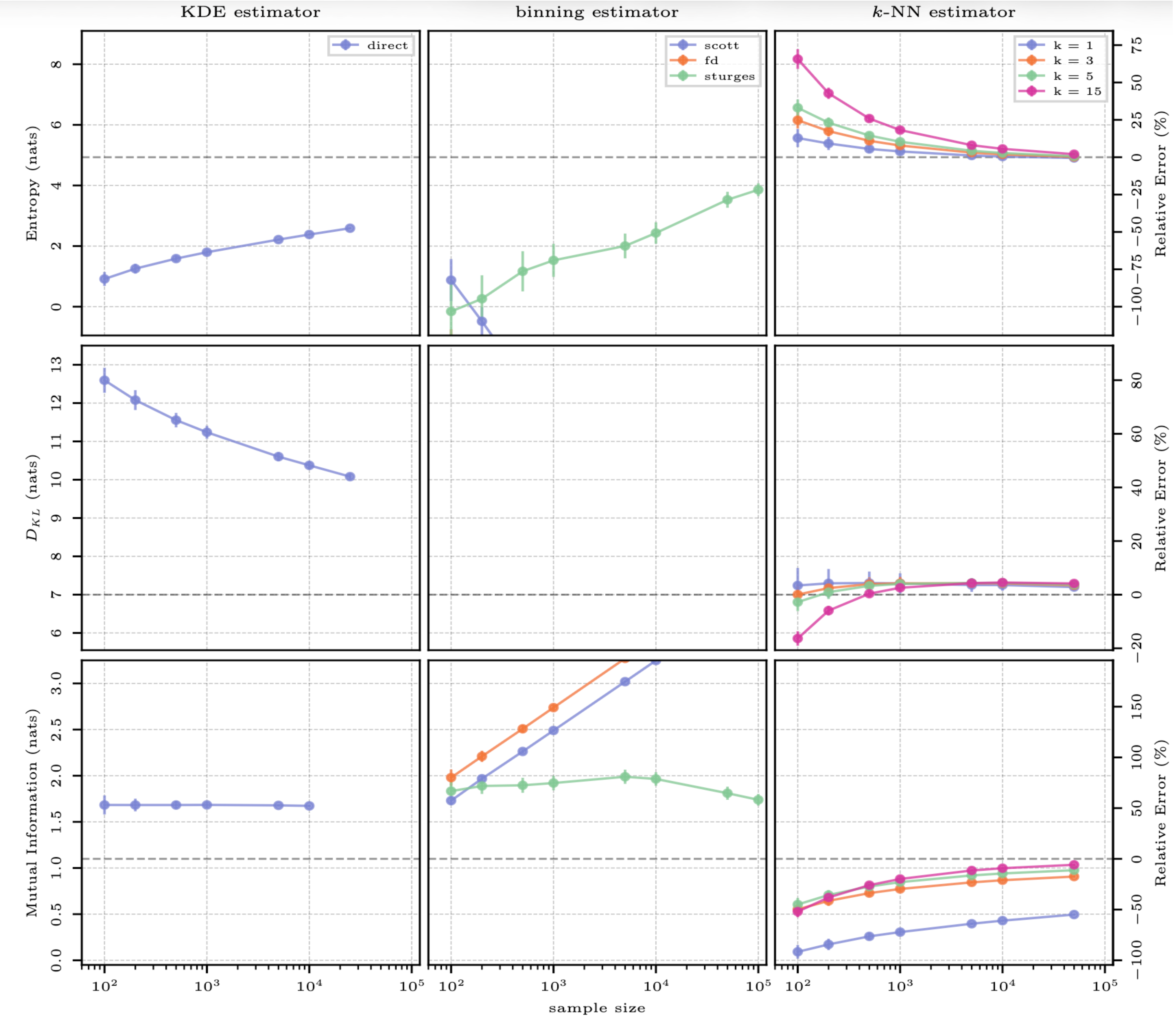
Álvarez Chaves et al. (2024), who test each estimator against **analytic ground** up to 10-D.

1. **k-NN converges for D_{KL}**

2. **KDE drifts, binning fails outright**

3. **⇒ No single estimator is trusted alone**
   So, we cross-check several methods — and keep every block at ≤5-D.

Álvarez Chaves, Gupta, Ehret & Guthke — “On the Accurate Estimation of Information-Theoretic Quantities from Multi-Dimensional Sample Data”, Entropy (2024).



Their Fig. 16 — 10-D Gaussian case: entropy (top), D_KL (middle), mutual information (bottom) vs. sample size for KDE, binning and k-NN estimators.

Group-KLD

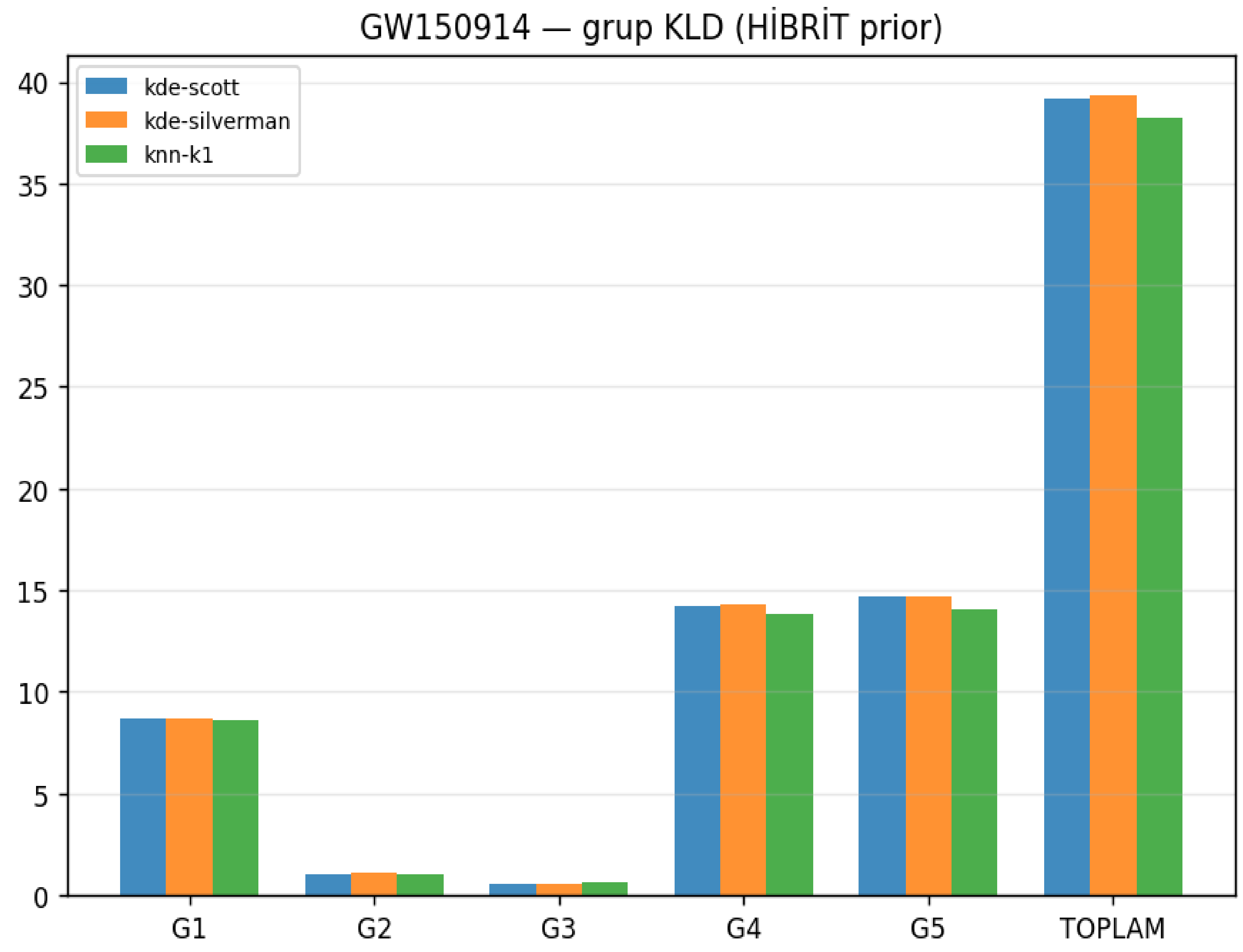
1 **Split 15-D into 5 physical blocks**

Masses · spins+tilts · spin phases · distance+geometry · sky+time — each ≤5-D, where density estimation is reliable.

2 **Cross-check three estimators**

KDE-Scott, KDE-Silverman and k-NN agree — the result is method-independent.

GW150914 → **≈ 38–39 bit** , consistent across estimators.



Per-group KL (left) and the marginal contributions (right) — three estimators overlap.

PART II

From One Event to a Population

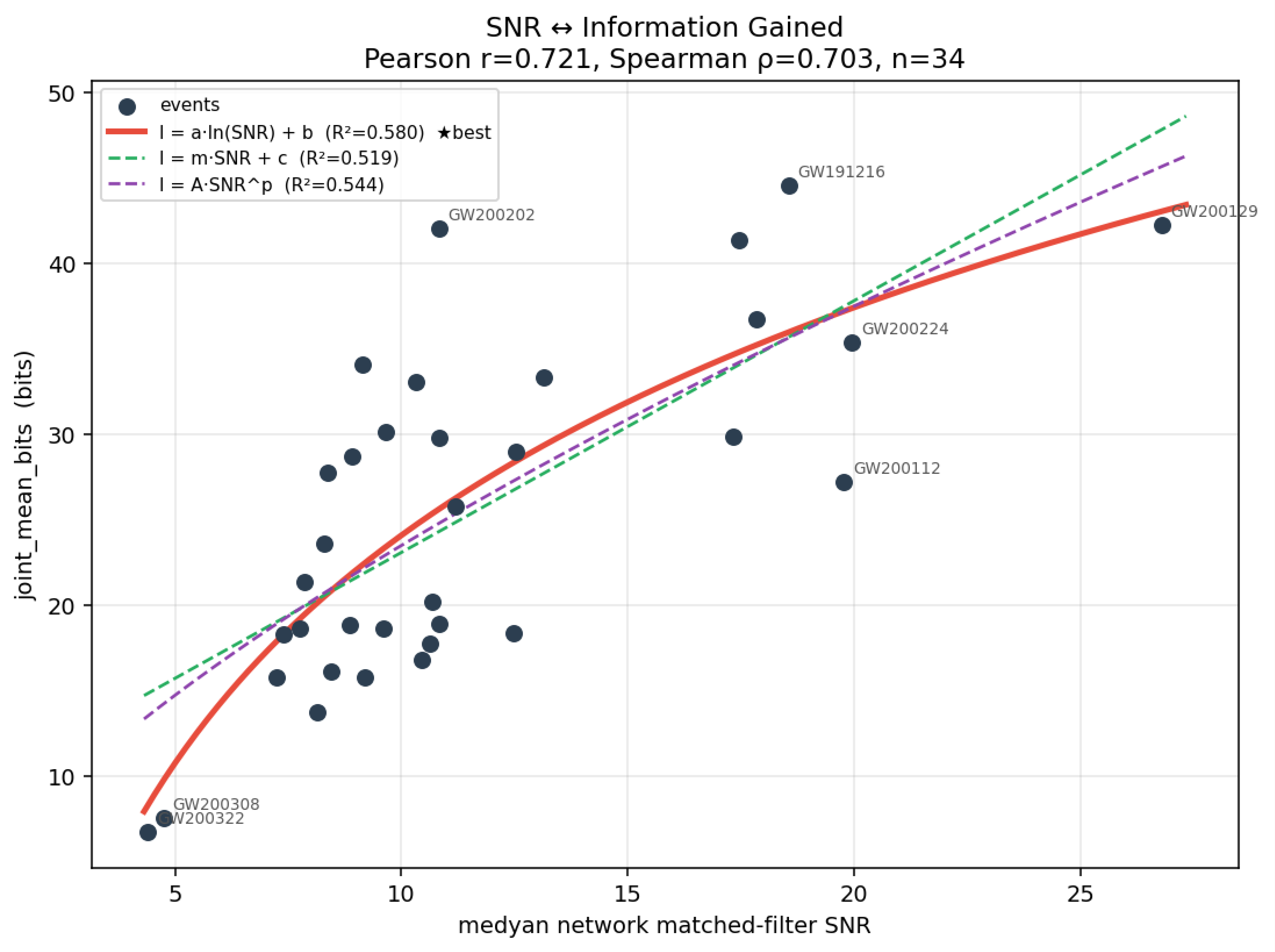
Applying the pipeline to all 35 events reveals a clean scaling law for information.

Information grows as the log of SNR

$$I = a \cdot \ln(\mathrm{SNR}) + b$$

0.58 R^2 **0.72** Pearson r

Across all 35 events the bits span **6.7 → 44.6** . The logarithm beats linear (0.52) and power (0.54) fits — exactly as expected when information scales like SNR^2.



Information vs. network SNR for all 35 events; three information measures track together.

Mass and detectors close the gap

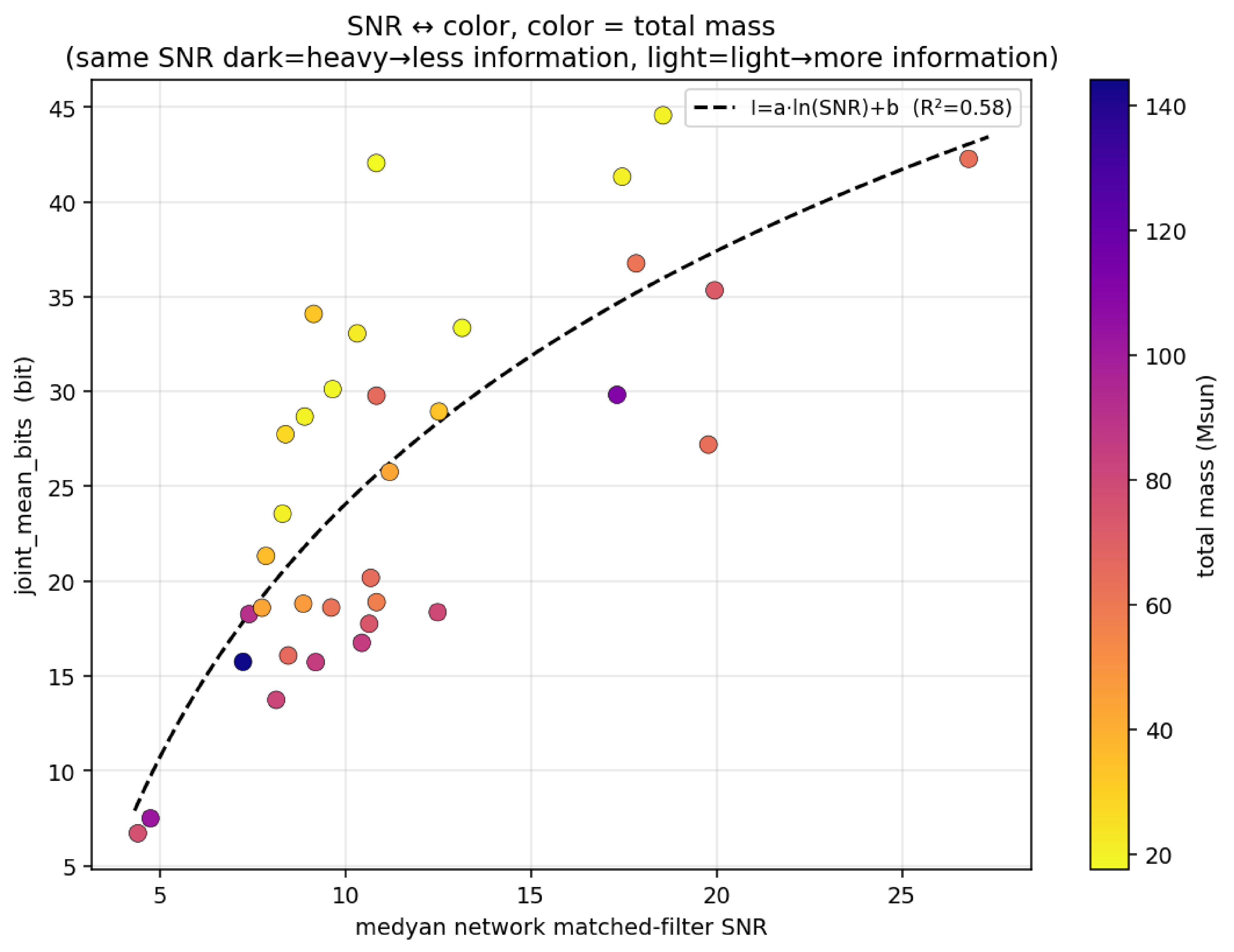
$$I = a\ln(\mathrm{SNR}) + c\ln(M) + d\,\mathrm{n}_{\mathrm{det}}$$

0.88 R^2

+ ln SNR louder → more information

− ln M heavier → **less** — light binaries ring for more cycles.

+ n_{det} more detectors → tighter sky & distance



Colour = total mass; SNR alone 0.58 → 0.88.

What we learned

01 A measurable quantity

Information per detection = D_{KL} (posterior‖prior), computed for real, non-Gaussian posteriors.

02 A method that scales

Group-KLD survives the curse of dimensionality and runs on all 35 events.

03 A scaling law

$I = a \cdot \ln(SNR) + c \cdot \ln(M) + d \cdot n_{det}$, $R^2 = 0.88$; louder & lighter mergers inform us most.

References — Flanagan & Hughes (1998) · Pérez-Cruz (2008) · Álvarez-Chaves et al. (2024) · Abbott et al., GWTC-3 (2023)

Thank you · questions?

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Thank you, do you have any questions?